Part 2 — Evolution and Refactoring

CH01 — Repository Pattern

Moved query/persistence operations from routes into EventRepository, TicketRepository and UserRepository. Pattern newly introduced; MVC and ORM collaborations modified. Structural tests: 4/4 passed.

CH02 — Strategy Pattern

Extracted event filtering criteria into TextSearchStrategy, CategoryFilterStrategy, DateFromFilterStrategy and DateToFilterStrategy. Strategy newly introduced. Structural tests: 4/4 passed.

CH03 — Observer Pattern

Added EventRegistrationService as subject plus notification and statistics observers. Registration route delegates registration and notification. Observer newly introduced. Structural tests: 4/4 passed.

Counts by snapshot: Original 4 types/4 instances; after CH01 5/5; after CH02 6/6; after CH03 7/7.

Metrics: routes.py 340 → 336 nonblank/non-comment LOC; new repository/strategy/observer/service modules add 193 such LOC. LOC movement is not treated as proof of quality.

Limitation: full Flask runtime was not claimed in the verification container because Flask dependencies were unavailable; the recorded structural/compilation evidence is the actual result.